

Bikendra

Senior Full Stack Engineer

SUMMARY

Full Stack Developer with 5+ years of experience building and maintaining scalable web and mobile applications using TypeScript, Node.js, Express.js, React Native, and MongoDB. Experienced in working with production-grade codebases, developing secure REST APIs, optimizing database performance, and delivering reliable features across the software development lifecycle. Strong understanding of cloud infrastructure, CI/CD pipelines, and third-party integrations, with hands-on experience deploying applications on AWS and Firebase ecosystems. An AI-native engineer who leverages modern AI tools to accelerate code comprehension, development, debugging, testing, and code reviews while maintaining high-quality engineering standards. Known for taking ownership of features end-to-end, collaborating effectively with cross-functional teams, solving complex technical challenges, and adapting quickly to evolving product requirements.

TECHNICAL EXPERTISE

- **Front-End Development:** React.js, React Native, Next.js, JavaScript (ES6+), TypeScript, HTML5, CSS3, Tailwind CSS, Styled Components, Responsive Web Design, Component-Based Architecture, React Hooks, Context API, Redux
- **Rendering & Performance:** Server-Side Rendering (SSR), Static Site Generation (SSG), Incremental Static Regeneration (ISR), Client-Side Rendering (CSR), Core Web Vitals Optimization
- **Back-End Development:** Node.js, Express.js, REST APIs, JWT Authentication, Background Workers, Cron Jobs, Atomic Operations, MongoDB Transactions, Webhooks
- **Database & ORM:** MongoDB, Mongoose, Prisma, Schema Design, Aggregation Pipelines, Indexing, Query Optimization, Database Migrations
- **Cloud & Infrastructure:** AWS S3, AWS Amplify, Amazon CloudFront, Google Cloud Platform (GCP), Firebase, Firebase Cloud Messaging (FCM), Google Vertex AI, Google Analytics (GA4)
- **Mobile Development:** React Native, Android, iOS, Google Play Console, Apple App Store Connect, In-App Purchases (IAP), Deep Linking, Remote Config
- **API Integration & External Services:** REST APIs, GraphQL, Apollo Client, Live Data APIs, Webhook Integration
- **Security & Authentication:** Authentication, Authorization, JWT, OAuth 2.0, Secrets Management
- **DevOps & Deployment:** GitLab CI/CD, CI/CD Pipelines, Deployment Automation, Environment Configuration, Production & Staging Management, Server Monitoring, Log Analysis, Incident Management, Sentry
- **Version Control & Collaboration:** Git, GitHub, GitLab
- **Testing & Quality Assurance:** Jest, React Testing Library, Cypress, Unit Testing, Integration Testing
- **Build Tools & Developer Tooling:** Webpack, Vite, Babel, npm, Yarn, ESLint, Prettier
- **AI Engineering & Developer Productivity:** AI-Assisted Development, AI Code Generation, AI Code Refactoring, AI-Assisted Debugging, AI Test Generation, Prompt Engineering, AI Code

Review, AI-Powered Architectural Analysis, AI-Assisted Production Code Navigation.

PROFESSIONAL EXPERIENCE

WitQualis | Senior Full Stack Engineer

2019 – Present

- Developed and maintained scalable full-stack applications using TypeScript, Node.js, Express.js, React Native, MongoDB (Mongoose), and REST APIs, implementing secure JWT authentication, role-based access control, and production-ready backend services while delivering high-quality features in an Agile environment.
- Designed and optimized MongoDB schemas, indexing strategies, aggregation pipelines, atomic operations, and transactions, implemented background workers, scheduled jobs, and webhook integrations with proper idempotency and distributed execution handling to ensure reliable, high-performance systems.
- Built, integrated, and deployed React Native mobile applications with Firebase (FCM, Remote Config, Deep Links), managed Android and iOS build pipelines, application signing, Play Store and App Store releases, and implemented In-App Purchase (IAP) workflows with receipt verification.
- Managed cloud infrastructure and deployment workflows using AWS (S3, Amplify, CloudFront), Google Cloud (Firebase, Vertex AI, GA4), and GitLab CI/CD, while handling environment configuration, secrets management, production monitoring, log analysis, incident triage, and error tracking with Sentry.
- Leveraged AI-native engineering practices by using AI-powered development tools for rapid production codebase navigation, intelligent code generation, refactoring, automated test creation, debugging, security reviews, architectural analysis, and documentation, significantly improving development efficiency while integrating external APIs, generative AI services, and client-side API contracts.

PROJECTS

Project 1: AI-Enabled Digital Audio Streaming & Subscription Platform

Role: Senior Software Engineer

Project Description

Developed and enhanced an enterprise-scale digital audiobook streaming platform supporting subscription management, content delivery, user authentication, personalized recommendations, and real-time media streaming. Built scalable backend services using Node.js, Express.js, TypeScript, MongoDB, and Mongoose, exposing secure RESTful APIs for content orchestration, subscription lifecycle, streaming metadata, and user management. Integrated Firebase Cloud Messaging (FCM) for push notifications, implemented JWT-based authentication, background workers, scheduled jobs, and cloud-native deployments on AWS (S3, CloudFront) with automated GitLab CI/CD pipelines. Leveraged AI-assisted development tools to accelerate production code navigation, debugging, API development, refactoring, and test generation, improving engineering productivity and release velocity.

Roles & Responsibilities

- Designed and developed scalable backend services using Node.js, Express.js, TypeScript, MongoDB, and Mongoose, implementing REST APIs, JWT authentication, MongoDB transactions, aggregation pipelines, scheduled jobs, and background workers for subscription management, audiobook streaming, and user activity workflows while optimizing query performance and application scalability.
- Integrated Firebase (FCM) notifications, AWS S3 and CloudFront for secure media delivery, implemented monitoring using Sentry, automated deployments through GitLab CI/CD, and utilized AI-powered development tools (Cursor AI, GitHub Copilot, Claude Code, ChatGPT) for code generation, debugging, refactoring, production issue analysis, and test automation.

Technologies & Tools Used

TypeScript, JavaScript (ES6+), Node.js, Express.js, MongoDB, Mongoose, Prisma, REST APIs, JWT Authentication, Firebase (FCM), AWS S3, CloudFront, GitLab CI/CD, Docker, Git, Sentry, Swagger/OpenAPI, Postman, Cron Jobs, Background Workers, MongoDB Transactions, Aggregation Pipelines, AI-assisted Development (Cursor AI, GitHub Copilot, Claude Code, ChatGPT), Agile, Jira.

Project 2: Enterprise Rental Marketplace & Reservation Platform

Role: Senior Software Engineer

Project Description

Developed a cloud-based rental marketplace enabling product listings, reservations, inventory synchronization, customer onboarding, payment workflows, and real-time booking management.

Engineered scalable backend APIs using Node.js, Express.js, TypeScript, MongoDB, and Mongoose, supporting high-volume transactional operations with MongoDB transactions and background workers. Integrated external payment and notification services, deployed applications on AWS, and leveraged AI-assisted engineering to improve development efficiency, code quality, and production troubleshooting.

Roles & Responsibilities

- Developed backend services for reservation processing, inventory synchronization, payment orchestration, customer lifecycle management, webhook integrations, and secure REST APIs using Node.js, Express.js, MongoDB, Mongoose, JWT Authentication, and optimized MongoDB indexing for improved performance.
- Automated deployments using GitLab CI/CD, implemented scheduled jobs, integrated Firebase Cloud Messaging, monitored production using Sentry, and utilized AI tools to accelerate feature implementation, debugging, refactoring, and architectural analysis across a distributed production environment.

Technologies & Tools Used

TypeScript, Node.js, Express.js, MongoDB, Mongoose, Prisma, REST APIs, JWT Authentication, Firebase (FCM), AWS S3, CloudFront, GitLab CI/CD, Docker, Git, Sentry, Swagger/OpenAPI, Postman, Webhooks, Cron Jobs, MongoDB Transactions, Aggregation Pipelines, AI-assisted Development (Cursor AI, Claude Code, GitHub Copilot, ChatGPT), Agile, Jira.

Project 3: AI-Powered Assessment & Evaluation Management Platform

Role: Senior Software Engineer

Project Description

Designed and enhanced an enterprise assessment platform supporting candidate onboarding, online examinations, automated evaluations, reporting, analytics, and AI-assisted content generation. Developed scalable backend services using Node.js, Express.js, TypeScript, MongoDB, and Mongoose, exposing secure REST APIs with JWT authentication. Integrated Google Vertex AI for AI-powered question generation and content workflows, automated deployments using GitLab CI/CD, and adopted AI-native engineering practices to accelerate production development and system optimization.

Roles & Responsibilities

- Built backend APIs for candidate management, assessment workflows, automated result processing, reporting, AI content generation, and third-party integrations using Node.js, Express.js, MongoDB, Mongoose, MongoDB Transactions, Aggregation Pipelines, and JWT authentication while ensuring scalable production performance.
- Integrated Google Vertex AI, Firebase notifications, background workers, scheduled jobs, Sentry monitoring, and GitLab CI/CD pipelines while leveraging AI coding assistants to improve debugging, code reviews, test generation, refactoring, and production issue resolution.

Technologies & Tools Used

TypeScript, Node.js, Express.js, MongoDB, Mongoose, Prisma, REST APIs, JWT Authentication, Google Vertex AI, Firebase (FCM), AWS S3, CloudFront, GitLab CI/CD, Docker, Git, Sentry, Swagger/OpenAPI, Postman, MongoDB Transactions, Aggregation Pipelines, Cron Jobs, AI-assisted Development (Cursor AI, Claude Code, GitHub Copilot, ChatGPT), Agile, Jira.

Project 4: Enterprise E-Commerce & Order Management Platform

Role: Senior Software Engineer

Project Description

Developed an enterprise e-commerce platform supporting product catalog management, customer orders, checkout, inventory synchronization, payment processing, and order fulfillment. Built scalable backend services using Node.js, Express.js, TypeScript, MongoDB, and Mongoose, exposing secure REST APIs with JWT authentication. Improved application scalability through cloud-native deployments on AWS, automated CI/CD pipelines, and AI-assisted development workflows for faster feature delivery and production support.

Roles & Responsibilities

- Designed backend APIs for product catalog, inventory, order lifecycle, checkout processing, webhook integrations, and customer management using Node.js, Express.js, MongoDB, Mongoose, implementing optimized aggregation pipelines and transactional operations for business-critical workflows.
- Implemented AWS-based deployments, GitLab CI/CD automation, Firebase notifications, Sentry monitoring, and AI-assisted development practices to accelerate development, production debugging, security reviews, and code optimization.

Technologies & Tools Used

TypeScript, JavaScript, Node.js, Express.js, MongoDB, Mongoose, Prisma, REST APIs, JWT Authentication, Firebase (FCM), AWS S3, CloudFront, GitLab CI/CD, Docker, Git, Sentry, Swagger/OpenAPI, Postman, MongoDB Transactions, Aggregation Pipelines, Webhooks, AI-assisted Development (Cursor AI, Claude Code, GitHub Copilot, ChatGPT), Agile, Jira.

Project 5: Enterprise Rental Operations & Workflow Automation Platform

Role: Senior Software Engineer

Project Description

Developed a cloud-native rental operations platform supporting booking orchestration, inventory management, workflow automation, reporting, customer lifecycle management, and third-party integrations. Designed scalable backend services using Node.js, Express.js, TypeScript, MongoDB, and Mongoose, implementing REST APIs, JWT authentication, scheduled jobs, and MongoDB transactions for state-sensitive business operations. Leveraged AWS infrastructure, GitLab CI/CD automation, and AI-native engineering practices to improve deployment efficiency, system reliability, and engineering productivity.

Roles & Responsibilities

- Designed and implemented backend services for rental workflows, booking management, reporting, inventory tracking, and external API integrations using Node.js, Express.js, MongoDB, Mongoose, leveraging aggregation pipelines, transactions, scheduled jobs, and optimized indexing for enterprise-scale performance.
- Automated deployments using GitLab CI/CD, integrated Firebase notifications, AWS cloud services, Sentry monitoring, and AI-powered development tools for production code analysis, debugging, automated testing, refactoring, and secure feature delivery.

Technologies & Tools Used

TypeScript, JavaScript (ES6+), Node.js, Express.js, MongoDB, Mongoose, Prisma, REST APIs, JWT Authentication, Firebase (FCM), AWS S3, CloudFront, GitLab CI/CD, Docker, Git, Sentry, Swagger/OpenAPI, Postman, MongoDB Transactions, Aggregation Pipelines, Cron Jobs, Background Workers, AI-assisted Development (Cursor AI, Claude Code, GitHub Copilot, ChatGPT), Agile, Jira.

EDUCATION

**Awadhesh Pratap Singh University
(APSU)** Bachelor of Engineering, Computer
Science Jun 2016 - Apr 2019

